Block GIT-207

Gastro-Intestinal System

Antiemetic Drugs/Anti-diarrheal/Laxatives /Spasmolytics/Prokinetics/Drug treatment of IBD & IBS

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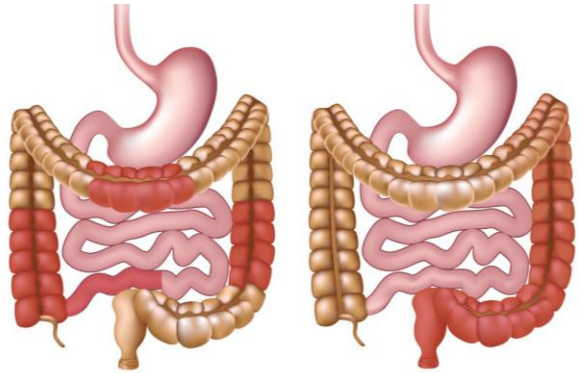

Antiemetic Drugs/Anti-diarrheal/Laxatives /Spasmolytics/Prokinetics/Drug treatment of IBD & IBS

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ILOs

- Pharmacology of antiemetic drugs.
- Pharmacology of antidiarrheal drugs.
- Pharmacology of laxatives.
- Pharmacology of antispasmodics (spasmolytics).
- Pharmacology of prokinetic drugs.
- Treatment of inflammatory Bowel disease (IBD).
- Treatment of irritable bowel syndrome (IBS).

Antiemetic Drugs

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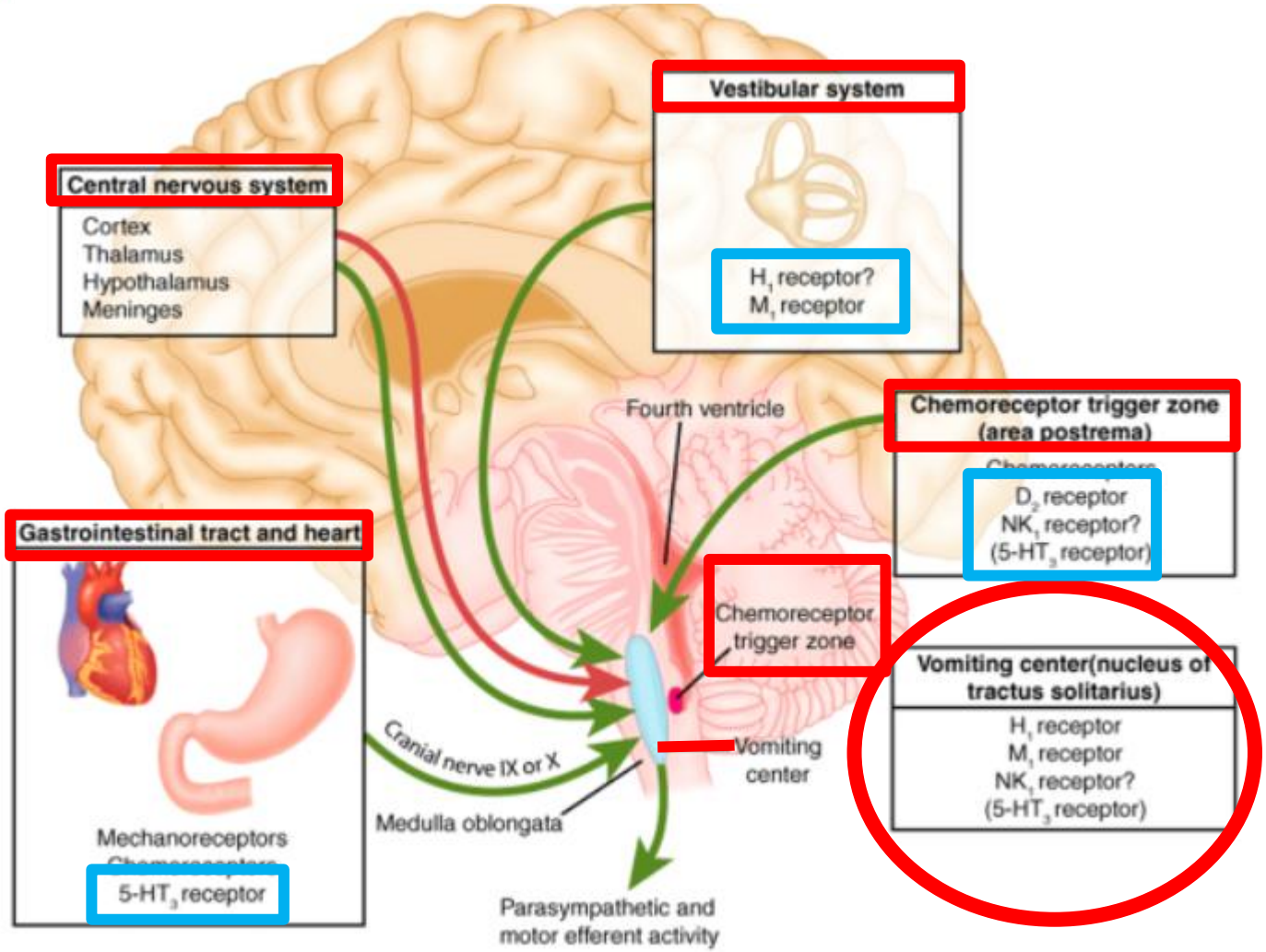
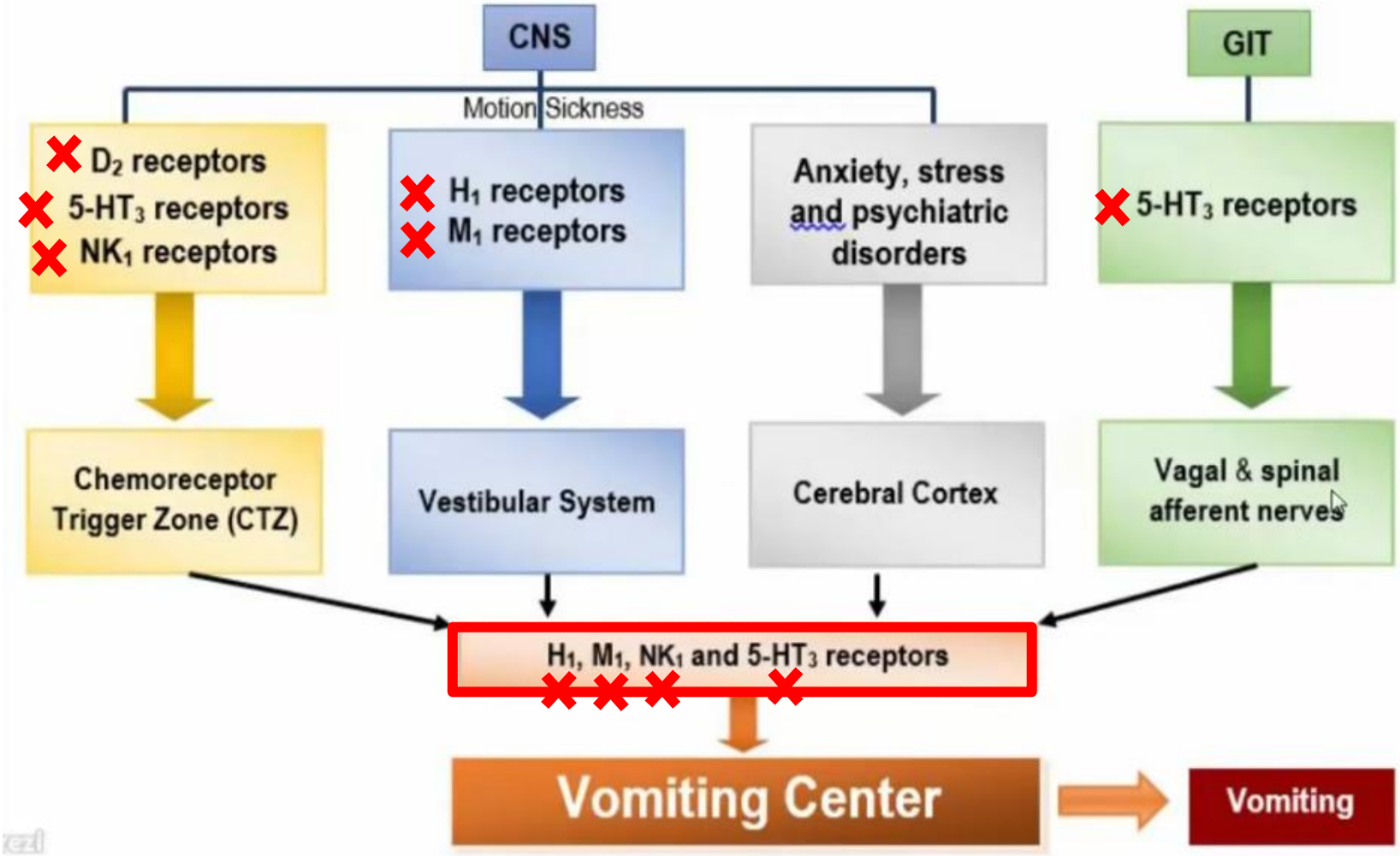
Antiemetic Drugs

- **Nausea: unpleasant sensation of imminent need to vomit** that may or may not lead to act of vomiting.
- **Vomiting: involuntary forceful expulsion of gastric contents through the mouth**.
- Nausea and vomiting occur in a **variety of conditions**:
  - Motion sickness,
  - Pregnancy, and
  - GIT illnesses,
  - Chemotherapeutic agents.

- The brainstem **vomiting center** is composed of a group of neuronal areas within the **medulla** that coordinate emesis. It may be stimulated by four different sources of afferent input:
  1. Afferent vagal fibers from **GIT viscera** (*rich in 5-HT3 receptors*).
  2. Fibers of the **vestibular system** (*rich in histamine H1 and muscarinic cholinergic receptors*).
  3. **Higher centers in brain** (amygdale area; where certain sights, smells or emotional experiences may induce vomiting)
  4. **The chemoreceptor trigger zone** (***CTZ***) which is *rich in opioid, serotonin 5-HT$_3$, neurokinin 1(NK1) and dopamine D$_2$ receptors*.


Vestibular system
H_1 receptor?
M_1 receptor
Central nervous system
Cortex
Thalamus
Hypothalamus
Meninges
Fourth ventricle
Chemoreceptor trigger zone (area postrema)
D_2 receptor
NK_1 receptor?
(5-HT_3 receptor)
Gastrointestinal tract and heart
Chemoreceptor trigger zone
Vomiting center(nucleus of tractus solitarius)
H_1 receptor
M_1 receptor
NK_1 receptor?
(5-HT_3 receptor)
Cranial nerve IX or X
Vomiting center
Medulla oblongata
Mechanoreceptors
5-HT_3 receptor
Parasympathetic and motor efferent activity


CNS
GIT
Motion Sickness
D2 receptors
5-HT3 receptors
NK1 receptors
H1 receptors
M1 receptors
Anxiety, stress and psychiatric disorders
5-HT3 receptors
Chemoreceptor Trigger Zone (CTZ)
Vestibular System
Cerebral Cortex
Vagal & spinal afferent nerves
H1, M1, NK1 and 5-HT3 receptors
Vomiting Center
Vomiting

- **Antiemetic drugs** are generally **more effective in prophylaxis than treatment**.
- Most antiemetic agents **relieve nausea and vomiting** by acting on the vomiting center, dopamine receptors, chemoreceptors trigger zone (CTZ), cerebral cortex, vestibular apparatus, or a combination of these.
- Drugs used in the treatment of nausea and vomiting belong to several different groups. These include:

Antiemetic Drugs

1. **Phenothiazines** such as chlorpromazine, act on CTZ and vomiting centre, block dopamine receptors, are effective in preventing or treating nausea and vomiting induced by drugs, radiation therapy, surgery and most other stimuli (e.g. pregnancy). They are generally **ineffective** in motion sickness.
2. **Antihistamines:** such as promethazine and dimenhyrinate are especially **effective** in prevention and treatment of motion sickness.
3. **Metoclopramide:** has both central and peripheral antiemetic effects. Centrally, it antagonizes the action of dopamine. Peripherally metoclopramide stimulates the release of acetylcholine, which in turn, increases the rate of gastric emptying. It has similar indications to those of chlorpromazine.
4. **Scopolamine:** an anticholinergic drug, is very effective in reliving nausea & vomiting **associated** with motion sickness.
5. **Ondansetron:** a serotonin antagonist at 5-HT_3, is effective in controlling *chemical-induced vomiting* and nausea such those induced by anticancer drugs.
6. **Aprepitant:** blocks neurokinin receptor (NK_1) to which substance P binds, it has significant antiemetic effect, it is indicated in vomiting during chemotherapy.

- "Royal combination" for resistant vomiting include: (***ondansetron + dexamethason + aprepitant***).

Anti-diarrheal Drugs

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- Diarrhea is a condition of having **at least three loose or liquid bowel movements in a day** **lasting for** **less than 14 days**.
- The most common cause of the gastroenteritis are due to either a **virus**, **bacteria**, or **parasite**.
- Diarrhea can be accompanied by **painful abdominal cramps**, nausea, fever, bloating, and generalized weakness.
- It can result in **dehydration** especially in young children and older people (**thirst, lack of energy, passing less urine than normal, dizziness or lightheadedness, and loss of skin color and turgor)**.
- Signs of **severe dehydration** in children include **sunken eyes**, **cheeks** or **belly** or **a sunken fontanelle**.

Antidiarrheal Drugs

Antidiarrheal drugs may be given **to relieve the symptom** (non-specific therapy) or to **treat the underlying cause of the symptom** (specific therapy).

Symptomatic (non-specific) Treatment	Specific therapy
1. Antimotility agents: diphenoxylate and **loperamide**. Both have opioid-like actions on the gut.	The use of **specific antibacterial agents** For example, severe diarrhea by salmonella, shigella, campylobacter and clostridia species can be treated by antibiotics (**ampicillin, chloramphenicol, co-trimoxazole**).
2. Adsorbents and demulcent: • **Aluminum hydroxide** and **methylcellulose** are used to control diarrhea. • These agents act by **adsorbing intestinal toxins or microorganisms** and/or by **coating or protecting the intestinal mucosa**. • They are **much less effective than antimotility agents**, and they **can interfere with the absorption of other drugs**.	
3. Agents that modify fluid and electrolyte transport: • **Bismuth subsalicylate**, used for **prevention and treatment of traveler's diarrhea**, **decreases fluid secretion** in the bowel. • Its action may be due to its **salicylate component** as well as its **coating action**. • It has **antimicrobial, bactericidal** action, **prevention of intestinal secretion and promotion of fluid absorption**.	

Laxatives

Constipation

- Constipation is defined as a **decrease in frequency and liquidity of stool compared to the normal pattern** in a particular individual.
- **Acute constipation** may be a part of a more serious illness such as acute bowel obstruction.

- The **definition** of **constipation** *includes* the ***following***;
  - * **Infrequent bowel movements** (*typically* ***less than*** 3 **times** ***per*** **week**).
  - * **Straining** ***during*** **bowel movements** (***difficulty*** *during* **defecation**).
  - * ***Sensation*** of **hard dry stools**.
  - * ***Sensation*** of ***incomplete*** **defecation**.

- **Patients** may **complain** ***one*** or ***more*** of the *following*:
  - * **Abdominal bloating**.
  - * **Pain on defecation**.
  - * **Rectal bleeding**.
  - * **Low back pain**.

Laxatives

- **Laxatives** and **cathartics** are drugs used orally to evacuate the bowels or to promote bowel elimination (defecation). Both terms are used interchangeably because it is the dose that determines the effects rather than a particular drug. For example, Castor oil laxative effect in a dose of 4ml while Cathartic effect in a dose of 15-60ml. The term laxative implies mild effects, and eliminative of soft formed stool. The term cathartic implies strong effects and elimination of liquid or semi liquid stool.

Laxatives

- **No drug measures to prevent constipation** (e.g increase intake of fluid and high fiber food, exercise).
- **Laxatives** are commonly used in the treatment of constipation
  - **accelerate the motility** of the bowel,
  - **soften the stool**, and
  - **increase the frequency of bowel movements**.
- **Laxatives**:
  - **Increase the potential for loss of pharmacologic effect of some oral preparations**.
  - Cause **electrolyte imbalances** when used chronically.
  - Many of these drugs have a risk of **dependency** for the user.
  - **Lifestyle advice, fluid intake, fiber, and exercise** must be continued throughout laxative therapy.

- Laxatives are randomly classified depending on mode of action as:

1. **Bulk-forming laxatives:** are substances that are largely unabsorbed from the intestine. They include psyllium, bran, methylcellulose, etc. When water is added, the substances swell and become gel-like which increases the bulk of the fecal mass that stimulates peristalsis and defecation.
2. **Osmotic laxatives:** such as magnesium sulphate, magnesium hydroxide, sodium phosphate, etc. These substances are not efficiently absorbed and cause water retention in the colon. The latter causes increase in volume and pressure which stimulates peristalsis and defecation.
3. **Stimulant (irritant) laxatives** these are irritant that stimulate elimination of large bowel contents. Individual drugs are castor oil, bisacodyl, phenolphthalein, cascara sagrada, glycerine, etc. The feces are moved too rapidly and watery stool is eliminated.
4. **Fecal softeners:** they decrease the surface tension of the fecal mass to allow water to penetrate into the stool. They have detergent- like property e.g. docusate.
5. **Lubricant laxatives** e.g. liquid paraffin (mineral oil). It lubricates the intestine and is thought to soften stool by preventing colonic absorption of fecal water. They are used as retention enema. Glycerin can be administered rectally as suppositories.

Clinical Uses

1. To **relieve constipation.**
2. To **prevent straining and piles.**
3. To **empty the bowel in preparation for bowel surgery or diagnostic procedures.**
4. To **accelerate elimination of potentially toxic substances from the GI tract.**
5. To **accelerate excretion of parasite after anti-helminthic drugs have been administered.**

Spasmolytic & prokinetic Drugs

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Spasmolytic Drugs

These are a heterogonous group of drugs, with **relaxing effect on the smooth muscle of GIT** and used for **symptomatic treatment** of acute and chronic diseases accompanied by **GIT smooth muscle hyper-activity**.

Parasympatolytics

a. Non-selective (general muscarinic receptor blockade – also bronchial, urogenital) like *atropine and scopolamine*, *butyl-scopolamine* (Buscopan) and the anti-asthmatic *ipratropium*.
b. Selective - specific M_1 blockers like *pirenzepine*.

Musculotropic spasmolytics

a) Papaverine-like type they have direct smooth muscle relaxation through increased cAMP (phosphodiesterase inhibition) like *papaverine* – opium alkaloid without analgesic effect and euphoria, it is also a vasodilator. *Drotaverin* is a strong spasmolytic and indicated in irritable bowel and biliary colic.

b) Calcium channel blockers *pinaverin* (Dicetel) is a selective inhibitor of calcium channel and indicated in irritable bowel and biliary colic.

Non-specific spasmolytics: e.g. nitrates, local anesthetics and aminophylline)

Adverse Reactions

- Parasympatolytics may cause **tachycardia, urine retention, excitation, increased intraocular pressure** or **precipitate glaucoma**.
- All others spasmolytic drugs and opioids can cause **paralytic ileus** and **toxic megacolon** if given in high doses.

Prokinetic Drugs

- Drugs that **enhance coordinated gastrointestinal motility and transit of materials downwards in the GIT**.
- Their prokinetic effect is **confined to upper GIT (lower esophagus, stomach and small intestine), but not the large intestine (colon).**
- They are used in treatment of **GERD** and **gastroparesis** (reduced gastric motility) associated with heart burn, nausea, vomiting and indigestion.

Metoclopramide: It has antiemetic effect, cholinomimetic action, and dopamine antagonistic action both central and peripheral levels. It is used for treatment of gastro-esophageal reflux and vomiting. It is cheap, fast, and has short time action (1-2 H).

Domperidone (Motilium) is a selective peripheral D2 receptor antagonist, it has antiemetic effect and it increases GIT motility and smooth muscle tonus, it has longer effect (4-6 H). It does not pass through the BBB; it has less adverse effects than metoclopramide but still causes hyper-prolactinemia present,

Cisapride: It acts directly as a 5-HT_4 receptor agonist which, stimulates Ach release from myenteric plexus in upper GIT to increase gastric motility, increase sphincter tonus, it has long t1/2 (10 H) . It has no antiemetic action; its adverse effects include: diarrhea, abdominal cramps, tachycardia and Q-T interval prolonged. It was withdrawn from market in USA.

Itoprid: It has dual action; peripheral D_2 receptor blockade and acetylcholinesterase inhibition. It is effective in treatment of GERD, gastroparesis and paralytic ileus. It is more expensive, highly effective and well tolerated than other prokinetics. It has quick onset and duration about (6-8 H). Adverse effects are less common and include: diarrhea, hyper-salivation and hyper-prolactinemia.

Irritable Bowel Syndrome (IBS)

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Irritable bowel syndrome (IBS)

- Irritable bowel syndrome (IBS) is characterized by
  - **chronic abdominal pain and**
  - **altered bowel habits** in the **absence of an organic cause**.
- IBS may be classified as
  - Constipation predominant (IBS-C),
  - (IBS-D), or Diarrhea predominant
  - a combination of both.
- **Diet** and **psychosocial modifications** play an important role in the management of the disease, as well as **drug therapy**.

Treatment

Dietary changes: Eliminating high-gas foods as carbonated beverages, vegetables especially cabbage, broccoli and cauliflower and raw fruits.

Medications:

- Fiber supplements. Taking fiber supplements, such as psyllium (Metamucil) or methylcellulose with fluids may help control constipation..
- Anti-diarrheal medications Over-the-counter medications, such as loperamide, can help control diarrhea. Some people will benefit from medications called bile acid binders, such as cholestyramine , colestipol , but these can lead to bloating.
- Anticholinergic and antispasmodic medications. These medications, such as hyoscyamine, can help relieve painful bowel spasms. They are sometimes used for people who have bouts of diarrhea, but they can worsen constipation and can lead to other symptoms, such as difficulty urinating.
- Antidepressant medications. If symptoms include pain or depression, a tricyclic antidepressant or a selective serotonin reuptake inhibitor (SSRI) could help. These medications help relieve depression as well as inhibit the activity of neurons that control the intestines.
- Antibiotics. Some people whose symptoms are due to an overgrowth of bacteria in their intestines may benefit from antibiotic treatment.

Medication specifically for IBS

Two medications are currently approved for specific cases of IBS:

1) **Alosetron "$5HT_3$ antagonists"**, appears to provide a modest benefit to women with IBS-D. It offers a new option for the treatment of IBS.
2) **Lubiprostone** works by increasing fluid secretion in your small intestine to help with the passage of stool. It is approved for women age 18 and older who have IBS with constipation. Its effectiveness in men is not proved, nor its long-term safety. Common side effects include nausea, diarrhea and abdominal pain. Lubiprostone is generally prescribed only for women with IBS and severe constipation for whom other treatments haven't been successful.

$5HT_4$ agonists and IBS

$5HT_4$ receptors present in the GI tract, they are G-protein-coupled receptors which mediate both relaxation and contraction of circular smooth-muscle strips and induce small bowel and to a lesser extent colonic fluid secretion, it Accelerate gastric emptying, accelerates small and large bowel transit and Increases stool water content

Tegaserod; it is a $5HT_4$ agonists, appears to be promising options in women with IBS not responding to increased dietary fiber or laxative therapy.

Inflammatory Bowel Disease (IBD)

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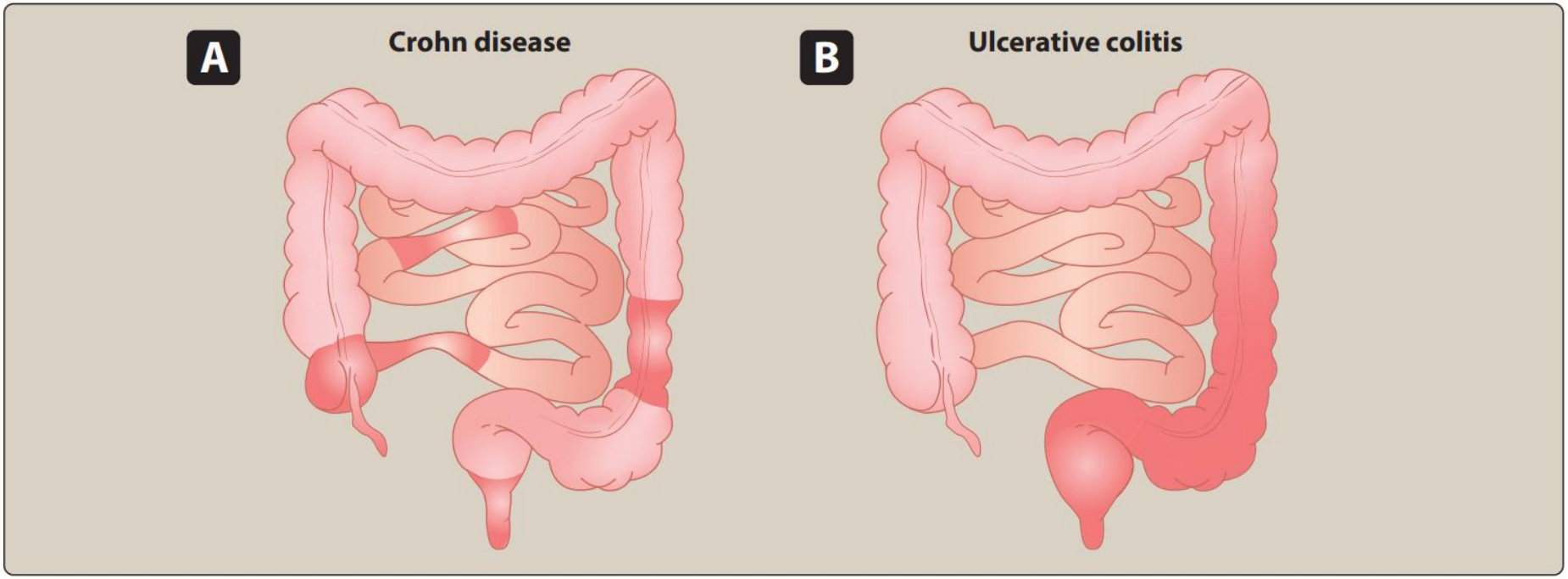
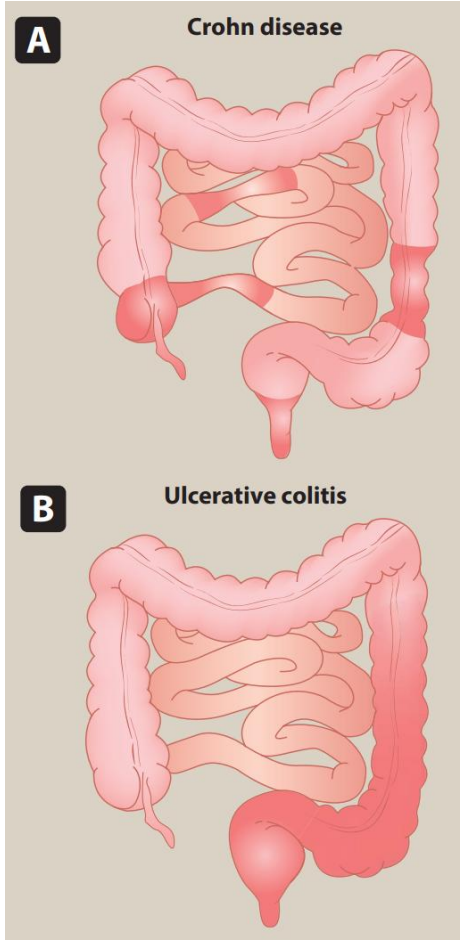
Inflammatory bowel disease (IBD)



- A group of idiopathic chronic intestinal conditions characterized by **immune-mediated GI tract inflammation in response to bacterial antigens in the intestinal lumen.**
- The most common subtypes are Crohn's disease (CD) and ulcerative colitis (UC).
- **CD** can **affect any portion of the GI tract from the mouth to the anus in a noncontinuous fashion** and is **characterized by transmural inflammation.**
- **UC** usually **affects the rectum.** It **may extend continuously to affect other parts of the colon and is characterized by inflammation limited to the mucosal layer.**



- **Remission** of IBD can be induced with the use of **rectal and oral 5-aminosalicylates (5-ASAs)**, **corticosteroids** (rectal, oral locally delivered and systemic), and **biologic agents** (TNF-α inhibitors, α4-integrin inhibitors, and the IL-12/23 inhibitor ustekinumab).
- The **immunomodulators** (**azathioprine, 6-mercaptopurine, and methotrexate**) are additional agents used in the maintenance of remission in IBD.

THANK
YOU